ic_autolab round 20 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T07:22:29Z at git `5958e283372a706805e09b8762d6c0554b28482b`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-b49545` (`python3 tools/pcc/pcc.py standing PCC-20261104-b49545`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k	class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
24	**improvement**	wsum-b8-n3	0.882 [0.839, 0.928]	0.892 [0.843, 0.944]	0.949 [0.867, 1.040]	sel < inc	3.678 / 3.920 / 6.100	4.170 / 4.395 / 6.425	1.209 / 1.318 / 2.301	3.0x / 3.0x / 2.7x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

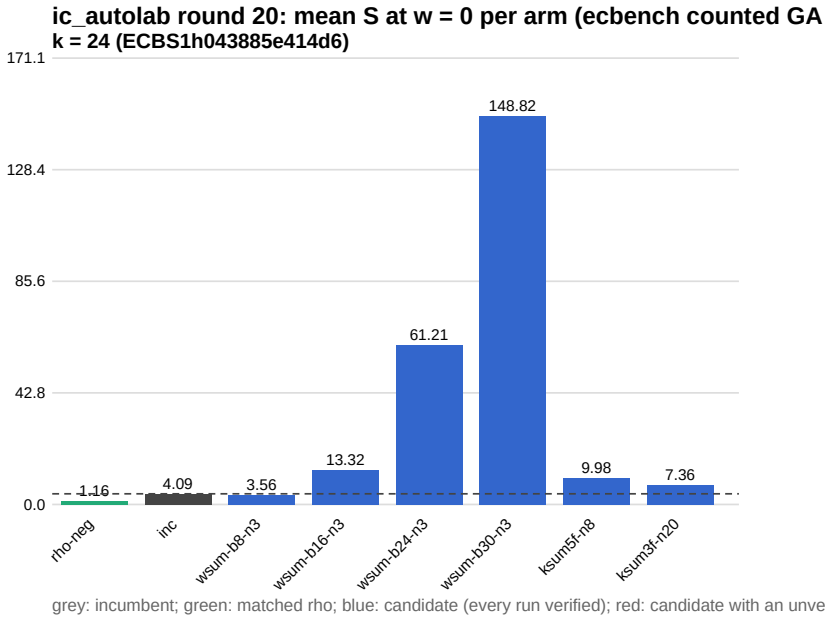
- k = 24: compare ops.status = ok, admissible = true; sel / inc = 0.8821 [0.839, 0.928] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1

Train (development screen, in sample)

k = 24: session `ECBS1h043885e414d6` (run exit 0, verify exit 0, receipt sha256 `e8b5381130f20cd890610458c12a1b8de87c0bf2a9d69e51865ca55ebcfd4511`)

arm	family	N factor	base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	-	-	`rho.negation`	32	32	1.162 / 1.266 / 2.207	matched reference
`inc`	wsum-b4	4	`prime-projected-abscissa:size=4,scan_cap=64`	`wsum:coef=4,free=1`	32	32	4.090 / 4.302 / 6.216	incumbent
`wsum-b8-n3`	wsum-b8	3	`prime-projected-abscissa:size=3,scan_cap=48`	`wsum:coef=8,free=1`	32	32	3.561 / 3.782 / 5.769	exploration: family wsum-b8 untried at k = 24, N = 0.0001 x N_a
`wsum-b16-n3`	wsum-b16	3	`prime-projected-abscissa:size=3,scan_cap=48`	`wsum:coef=16,free=1`	32	32	13.318 / 13.352 / 13.667	exploration: family wsum-b16 untried at k = 24, N = 0.0001 x N_a
`wsum-b24-n3`	wsum-b24	3	`prime-projected-abscissa:size=3,scan_cap=48`	`wsum:coef=24,free=1`	32	32	61.210 / 63.223 / 81.345	exploration: family wsum-b24 untried at k = 24, N = 0.0001 x N_a

arm	family	N	factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
wsum-b30-n3	wsum-b30	3	prime-projected-abscissa:size=3,scan_cap=48	wsum:coef=30,free=1	32	32	148.818 / 155.964 / 220.277	exploration: family wsum-b30 untried at k = 24, N = 0.0001 x N_a
ksum5f-n8	ksum5f	8	prime-projected-abscissa:size=8,scan_cap=128	ksum:m=5,free=1	32	32	9.980 / 11.463 / 24.816	exploration: family ksum5f untried at k = 24, at its parent ksum5's best N = 6
ksum3f-n20	ksum3f	20	prime-projected-abscissa:size=20,scan_cap=320	ksum:m=3,free=1	32	32	7.358 / 7.954 / 13.316	exploration: family ksum3f untried at k = 24, at its parent ksum3's best N = 20



train screen

Selection

- k = 24: challenger `wsum-b8-n3` (Pareto alternative at w = 1: `wsum-b8-n3`)

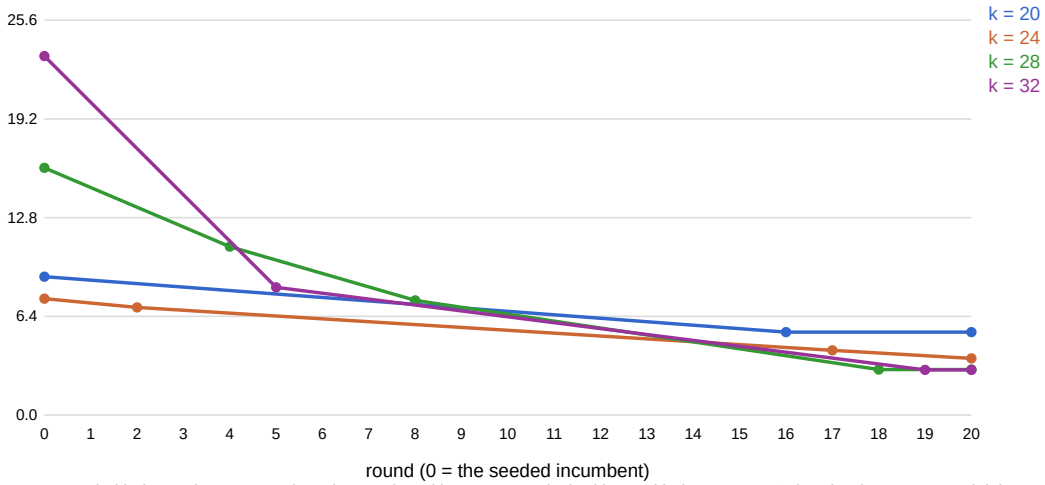
Holdout sessions

k	session	run exit	verify exit	receipt sha256	levels
24	ECBS1hbaf914946ee3	0	0	5802fde6e6d45085d71c4b903ab3a674fb0a619f2d2753b9475b016d6e4a6fe4	{"verified":480}

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	`wsum-b5-n3` (`prime-projected-abscissa:size=3,scan_cap=48` + `wsum:coef=5,free=1`)	16	5.384 / 5.609 / 7.638	3.6x / 3.5x / 3.0x
24	`wsum-b8-n3` (`prime-projected-abscissa:size=3,scan_cap=48` + `wsum:coef=8,free=1`)	20	3.678 / 3.920 / 6.100	3.0x / 3.0x / 2.7x
28	`wsum-b12-n3` (`prime-projected-abscissa:size=3,scan_cap=48` + `wsum:coef=12,free=1`)	18	2.949 / 3.222 / 5.676	3.0x / 3.0x / 3.0x
32	`wsum-b20-n3` (`prime-projected-abscissa:size=3,scan_cap=48` + `wsum:coef=20,free=1`)	19	2.937 / 3.213 / 5.697	3.1x / 3.1x / 3.1x

Ratchet: incumbent mean S at w = 0 by round (generic track, Curve25519 miniatures)



Recorded ledger values; a step down is a ratcheted improvement (paired interval below 1 at w = 0, band order at every weight).

ratchet

Evidence and deviations

- ecbench binary sha256: `e7242bc192bda8c8da9e64895afe943e8aa47cfa37bd490d656e2181758e9533`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.